Cleaning Log - customer_churn

Rows before cleaning 10,300
Rows after cleaning 10,000

#	Step	Column	Issue	Count	Decision	Reason
1	Remove Duplicates	All columns	Duplicate rows (same Customer_ID)	300	Removed duplicate rows	The same customer was counted twice, which infalates the totals and churn rate
2	Standardize Text	Region	Inconsistent casing (south, west, east, north)	1,001	Unified all values to South, West, East, North.	SQL, Python treat South and south as different values and would split one region in two.
3	Standardize Text	Subscription_Plan	Inconsistent casing (BASIC, PRO, PLUS)	1,035	Unified all values to Basic, Pro, Plus.	Same reason of 3, one plan would appear as 2 or more different groups
4	Invalid -> null	Age	Impossible ages (negative: 13, zero: 6, 150: 10)	29	Converted to null	Ages like -5 or 150 are impossible, so they are entry error. I kept the rest of the rows because the data is valid.
5	Invalid -> null	Age	Ages under 13 (1 to 12)	148	Converted to null	A paying subscriber under 13 is unlikely, so I treated these as errors. This cutoff is my own judgment.
6	Missing Values	Age	Originally blank	995	Kept as null, then filled with the median (row 12)	Deleting these rows would lose about 10% of customers, and nulls cannot be used in calculations.
7	Invalid -> null	Customer_Service_Calls	Negative value (-1)	50	Converted to null	A negative number of calls is impossible.
8	Invalid -> null	Monthly_Spend	Negative value (-10)	48	Converted to null	Spend cannot be negative, and -10 looks like a system placeholder.
9	Invalid -> null	Monthly_Spend	Placeholder value (99999.99)	11	Converted to null	99999.99 is far above real values (the maximum is about 311). It would raise the average spend from about 76 to about 191
10	Missing Values	Monthly_Spend	Originally blank	500	Kept as null, then filled with the median (row 14)	Spend depends on the plan, so a single overall median would be wrong.
11	Missing -> Unknown	Payment_Method	Blank values	700	Converted to unknown	Payment method is a category, so a median is not possible. Unknown avoids guessing and keeps these customers in the analysis.
12	Median Imputation	Age	Total nulls filled with 34 (rows 4 + 5 + 6)	1,172	Filled nulls with the median (34)	The median is not affected by extreme values. Here the mean (33.9) and median (34) are almost the same, so the effect is small.
13	Median Imputation	Customer_Service_Calls	Nulls filled with 2	50	Filled nulls with the median (2)	Calls are whole numbers. The median (2) is a whole number, but the mean (1.81) is not.
14	Group-wise Median Imputation	Monthly_Spend	Total nulls filled (rows 8 + 9 + 10): Basic 268 with 29.62, Pro 183 with 85.075, Plus 108 with 159.645	559	Filled nulls with the median of each plan	Spend differs a lot by plan (Basic 29.6, Pro 85, Plus 159.6), so one overall median (66.6) would be wrong for most customers

Observations (left unchanged)

#	Column	Observation	Count	Decision	Reason
1	Monthly_Spend	Value of 0 (mostly Basic plan)	1,078	Kept as it is	It could be a free tier or inactive customers. I would ask the business owner before changing it.
2	Gender	Value Unknown	961	Kept as a separate category	Guessing gender would add made-up data.
3	Gender	Value Non-Binary	1,001	Kept	It is a valid category, not an error.
4	Days_Since_Last_Login, Churn	No issues found	0	No action	I checked them, and there were no missing or invalid values.